



LabVIEW 7.1 front panel (Courtesy: commons.wikimedia.org)

ecosystem around their software and hardware platforms.

Advantages for engineers using the software-centric approach

The advantages of possessing a software-centric ecosystem for your hardware needs are many. From immediate benefits like increasing productivity and optimising costs to the long-term benefits like being able to go into previously niche areas, the adoption of a software-led ecosystem is going to be crucial in the coming years.

Productivity. With software defined instrumentation, and with software IPs available easily from the ecosystem, the test engineer has the freedom to quickly mix and match various commercial off-the-shelf (COTS) hardware and software to customise a generic test system to test multiple products. This drastically increases the productivity of the test engineer, which is very much required in today's scenario where more and more products get released every day.

Cost optimisation. Operational test costs are a combination of development costs, deployment costs and operational costs. A platform and ecosystem based approach, and hardware and software reuse for multiple tests, reduces development costs because of lower hardware requirement and the reuse of hardware for different test set-ups. This also reduces operational costs because of lower space require-

ments, as modular instruments have a compact form factor, and cuts deployment costs by driving down the man hours required for tests and reducing the need of specialised training on different tools.

Penetration into niche areas. An ecosystem makes it possible for domain experts to expand the power of the hardware and software platforms of a T&M player by developing and sharing add-ons and toolkits for applications in their domains. This provides an opportunity to the T&M player to expand into niche domains, which may not have been its strong point, earlier.

Madhukar Tripathi, Regional Manager, Anritsu India states, "Future measurement techniques will be based on software or T&M based on software only. The utility of any T&M instrument is based purely on its software, Graphical User Interface (GUI). An easy GUI makes it more user-friendly and saves a lot of testing time. Online help, remote control, automation/automatic measurement are some trends which, we believe, will dominate T&M ecosystem."

Patrick Allen, Sci-Mech Technical Services, says, "Once, we had to design and develop an integrated and reliable test system to control test parameters, such as drive speed and oil temperatures, while acquiring data from various sources such as thermocouples, accelerometers, pressure transducers, speed sensors, flow meters and transmission-specific sensors. We

were at our wits' end. Thankfully, using LabVIEW software with SCXI and PXI hardware, we were able to create a data acquisition and control system for aircraft gearbox testing that not only met the project's requirements but had the flexibility for future expansion."

The challenges faced

A software-centric ecosystem can be successful only if the members of the ecosystem are capable of developing good-quality IP, and are willing to share their IP with their peers. To ensure good-quality-content creation within the community, it becomes imperative for the T&M players to train community members on their hardware and software platforms, and educate them about the best practices and conventions to follow while writing add-ons, toolkits, etc. Since an ecosystem adds value to a T&M player's proposition to the market, and improves its customers' productivity, it is important for that player to facilitate such training. For instance, free webcasts, tutorials, best practices documents, self-paced training, etc, are available to educate and encourage community members to develop and share IP, and increase add-ons to its software and hardware portfolio.

Sambit Panigrahi, Texas Instruments, adds, "We were once faced with the unenviable task of having to develop a modular test solution that was abstract, scalable and easy to use, while supporting test sequencing across hundreds of power management ICs (PMIC) and interacting with multiple instruments, evaluation modules and source measure units (SMUs). Luckily, we were able to utilise LabVIEW to build a flexible and modular automated test solution that could test several PMICs with different requirements, communications buses and protocols."

An ideal scenario for T&M players would be that community members share their IP free of cost with everyone. This, however, is not possible in most cases, and T&M players have to deal with community members whose employers do not allow them to share